

# Waqas Naveed

Data Engineer | Cloud & Real-Time Data Engineering | Big Data & Analytics

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## PROFILE

Data Engineer with hands-on experience in real-time data streaming, cloud data platforms, ETL/ELT, distributed processing, data lakes, data warehousing and OLAP systems. Build scalable pipelines with **Python, SQL, Apache Kafka, Apache Spark, Databricks, Scala, Apache Airflow, AWS and Azure** — including a real-time multi-cloud platform processing approximately **1,000 transactional records per second** across streaming, orchestration, Delta Lake and real-time analytics. Also work across analytics and BI reporting (**Pandas, Power BI, Tableau**), with additional background in AI/ML, backend development and distributed systems.

## WORK EXPERIENCE

### ASTRA Innovation, AI/ML Engineer

08/2025 – 08/2026 | Pakistan

- Developed Python-based AI/ML and data-processing systems, and built AI-powered workflows spanning data processing, machine learning and automation.
- Built and integrated APIs, databases and scalable application components within production-oriented cloud and backend architectures.

### KryptoMind, AI Intern

06/2025 – 08/2025 | Pakistan

- Worked on Python-based AI and machine-learning workflows, including data preparation and processing for ML applications.
- Assisted with model development, testing and integration to deliver practical AI solutions across data and application workflows.

## EDUCATION

BS Computer Science, National University of Computer and Emerging Sciences (FAST-NUCES), Pakistan

2021 – 2025

## TECHNICAL SKILLS

### Programming

Python | SQL | Scala | Bash | JavaScript / TypeScript

### Data Engineering

ETL / ELT | Batch & Real-Time Pipelines | Data Modeling | Data Warehousing | Data Lakes | OLTP / OLAP | Star & Snowflake Schema | Fact & Dimension Tables | Bronze–Silver–Gold | Data Quality & Lineage

### Big Data & Streaming

Apache Kafka (Producers, Consumers, Topics, Partitions, Consumer Groups) | Amazon MSK | Azure Event Hubs | Apache Spark | PySpark | Spark Structured Streaming | Databricks | Delta Lake | Event-Driven Architecture

### Databases

PostgreSQL | PostGIS | MySQL | MongoDB | Redis (GEO, Pub/Sub) | Vector Databases

### Data Analysis & Visualization

Pandas | NumPy | Matplotlib | Seaborn | Power BI | Tableau

### AWS

S3 | Redshift | MSK | Glue | Athena | Aurora PostgreSQL | ECS / Fargate | ElastiCache / Redis | SQS | EventBridge | IAM | KMS | Secrets Manager | CloudWatch | CloudFront | Route 53 | WAF

### Azure

Data Lake Storage Gen2 | Data Factory | Databricks | Synapse Analytics | Event Hubs | Azure Storage | IAM / RBAC | Azure CLI

### Orchestration & DevOps

Apache Airflow | dbt | Docker | Kubernetes | Terraform | Git | GitHub | CI/CD | Linux | AWS CLI | Azure CLI

### Backend & Distributed Systems

Node.js | FastAPI | Flask | REST APIs | WebSockets | Redis Pub/Sub | Transactional Outbox | Fault Tolerance | Horizontal Scaling | Real-Time Systems

### AI / Machine Learning

Machine Learning | Deep Learning | NLP | Computer Vision | Generative AI | LLMs | RAG | AI Agents | LangChain | OpenAI | Claude | Gemini | Hugging Face | TensorFlow | PyTorch | Scikit-learn

### Frontend

React.js | React Native | TypeScript | Vite | Responsive UI | Real-Time Dashboards

## PROJECTS

### Real-Time Multi-Cloud Data Engineering Platform

AWS | Azure | Kafka | Databricks | Scala | Spark | Airflow | Delta Lake

- Built a real-time platform generating and processing approximately **1,000 transactional records per second**, with continuous ingestion and distributed processing across a multi-cloud **AWS + Azure** architecture.
- Implemented event streaming with **Apache Kafka** and cloud streaming infrastructure; processed streams in **Azure Databricks** using **Scala** and **Apache Spark**.
- Structured **Bronze → Silver → Gold** layers on **Delta Lake**, automated ingestion and transformation with **Apache Airflow**, and surfaced real-time analytics through a **React.js** monitoring dashboard.

### SastiRide, Real-Time Ride-Hailing & Courier Platform

AWS | Node.js | PostgreSQL | PostGIS | Redis | WebSockets | Kafka

- Designed and developed a scalable ride-sharing and courier platform for Pakistan covering customer, driver, courier and administrator workflows.
- Implemented real-time driver location tracking and ride matching using **Redis GEO** and **WebSockets**, with **PostgreSQL/PostGIS** as the transactional layer and Redis for hot state, caching and coordination.
- Built event-driven processing via **Kafka** and a transactional outbox, plus KYC/vehicle-document verification, secure object storage, authentication and audit logging; delivered **React Native** apps for Android and iOS alongside a responsive React admin platform.

### School Analytics Data Pipeline

PostgreSQL | Python | AWS S3 | PySpark | Redshift | dbt | Airflow

- Built an end-to-end **OLTP → Data Lake → Spark → OLAP** pipeline: Python ETL from PostgreSQL into **Amazon S3**, transformation in **PySpark**, and fact/dimension models loaded into **Amazon Redshift**.
- Modeled analytical transformations with **dbt** and automated end-to-end execution with **Apache Airflow**.

### Azure IoT Manufacturing Analytics

Azure Data Factory | ADLS Gen2 | Databricks | Delta Lake | Synapse

- Designed a cloud analytics architecture for continuously arriving IoT data, ingesting and orchestrating with **Azure Data Factory** into **ADLS Gen2**.
- Processed data in **Azure Databricks** and **Spark** across Bronze, Silver and Gold **Delta Lake** layers, and modeled OLAP-oriented analytics for **Azure Synapse Analytics**.